

Rowan Gaertner

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Education

University of Waterloo

Sep 2023 – Present

Bachelor of Software Engineering, Honours

Waterloo, ON

- Coursework: Algorithms, Data Structures, Operating Systems, Compilers, Computer Architecture, Databases
- Alumni Scholarship, President's Scholarship, Software Engineering Society Representative, Yearbook Rep

Experience

Transify

Sep 2025 – Dec 2025

Software Engineering Intern

Toronto, ON

- Created data visualization wizard in React, enabling **24 transit agencies** to self-serve reports without training
- Built flex route service area interpolation in TypeScript and Turf.js, expanding coverage from **30% to full route**
- Shipped trip filtering tools with drawable polygons and deck.gl arc layers for origin-destination analysis
- Designed **async batch processing** for route geometry inference, reducing per-segment API calls to one per route

Miovision

Jan 2025 – Apr 2025

Software Developer Intern

Waterloo, ON

- Delivered **lane separation** feature in Angular and ASP.NET, enabling per-lane traffic analysis in study workflows
- Extended client-facing report generation in C#, with configurable speed metrics, custom branding, and CSV exports
- Investigated DST de-sync in ATR timestamp processing; presented tradeoff recommendation to leadership

North Arrow Research

May 2024 – Aug 2024

Software Engineer Intern

Vancouver, BC

- Overhauled Python templates used to generate **110,240** HTML watershed reports with improved layout and charts
- Automated media pipeline with Node.js and TypeScript to generate timelapse GIFs, improving rich link SEO
- Developed SQL and Python scripts generating analytics reports for **86,000+** projects on web and mobile apps

PixelPAD.io

Jun 2021 – Aug 2023

Software Developer

Vancouver, BC (Remote)

- Diagnosed eager-loading bottleneck in notification system; implemented lazy fetch to **reduce DB queries by 55%**
- Created Cypress integration/E2E testing suite from scratch, reaching **82% coverage** across 870 components
- Engineered visual game development environment that **transpiles** drag-and-drop to Python for new programmers

Projects

RepoStory (Hack the North 2025 Winner) | TypeScript, Python, FastAPI, Next.js

- Built **AI agent** to analyze GitHub repositories and generate readable milestone timelines from commit history
- Optimized git operations by fetching only partial tree objects and using git bisect with custom **heuristics**
- Designed lazy file diff retrieval using **OpenAI Agents SDK**, minimizing context window usage while streaming results

Automatic Laser Pointer | Python, OpenCV, MediaPipe, Raspberry Pi

- Created a device that automatically aims and shines a laser pointer as directed by hand gestures
- Integrated **OpenCV** and **MediaPipe** on a local device, communicated servo targets to Raspberry Pi via sockets

Platform Wars – Hackable web game | Node.js, Python, Express, Socket.IO

- Built a hackable multiplayer game with **14,000+ plays**; users can modify local source code while synced with others
- Deployed general-purpose multiplayer API using Express and Socket.IO to enable real-time web games

Technical Skills

- **Languages:** Python, C/C++, TypeScript, SQL, Scala, Bash, C#, Assembly, GraphQL, PHP, HTML/CSS
- **Technologies:** React, Django, TensorFlow, NumPy, Express, Angular, .NET, React Native, Cypress, Jest
- **Tools:** AWS, GCP, Docker, GDB, Make, Git, CI/CD, Valgrind, Jupyter, DataDog, Figma, Claude Code, Arduino